

LISBOA GAGO COUTINHO, Portugal

WMO: 085790

Lat: 38.7662N

Lon: 9.1275W

Elev: 105

StdP: 100.07

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	4.7	5.9	-3.4	2.9	8.6	-1.3	3.4	9.4	10.0	13.9	8.9	12.9	2.9	50	0.408

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	10.1	33.7	20.1	31.7	19.6	29.7	19.2	21.5	30.0	20.8	28.2	20.2	26.8	3.1	330

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.5	14.4	22.2	18.7	13.7	22.1	18.1	13.2	21.9	62.8	30.1	60.4	28.3	58.3	26.8	24.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.3	7.3	6.4	2.4	37.7	1.3	2.3	1.4	39.4	0.7	40.7	-0.1	42.0	-1.0	43.7		
			0.5	23.2	1.6	0.7	-0.7	23.7	-1.6	24.2	-2.5	24.6	-3.6	25.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	17.2	11.3	12.0	14.3	15.7	18.1	21.2	22.7	23.5	21.9	18.9	14.4	11.9
	DBStd	5.01	2.36	2.10	2.48	2.83	3.10	3.04	2.70	2.64	2.46	2.45	2.44	2.40
	HDD10.0	33	14	7	1	0	0	0	0	0	0	0	1	10
	HDD18.3	999	218	176	127	89	41	5	0	0	1	22	119	201
	CDD10.0	2655	54	64	134	171	252	335	394	418	357	276	133	67
	CDD18.3	581	0	0	2	11	35	91	136	159	107	39	2	0
	CDH23.3	3732	0	0	13	60	244	618	950	1132	573	142	1	0
	CDH26.7	1318	0	0	1	7	74	229	359	437	182	29	0	0
Wind	WSAvg	3.2	2.9	3.1	3.3	3.3	3.4	3.4	3.6	3.3	2.9	2.9	2.8	3.0
Precipitation	PrecAvg	735	105	92	72	62	47	18	3	5	30	94	121	101
	PrecMax	1189	353	298	267	163	156	98	43	58	129	262	373	383
	PrecMin	416	0	0	1	5	0	0	0	0	0	0	2	4
	PrecStd	206	76	67	59	40	40	22	7	11	32	74	86	91
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.7	19.5	24.9	27.7	31.9	34.9	36.2	36.6	33.4	29.8	22.3	18.1
		MCWB	13.9	13.4	14.5	17.1	19.3	20.5	21.3	20.5	19.4	17.3	16.7	15.6
	2%	DB	16.6	17.7	22.7	25.0	28.7	31.7	33.2	33.7	31.0	26.7	20.4	17.2
		MCWB	13.9	12.6	14.1	16.1	18.0	19.9	20.2	20.3	18.9	17.5	16.5	15.4
	5%	DB	15.7	16.5	20.7	22.8	26.1	29.3	30.5	31.2	28.8	24.6	19.2	16.5
		MCWB	13.5	12.6	13.6	15.3	17.1	19.0	19.8	19.7	18.8	17.2	16.0	14.8
	10%	DB	14.9	15.5	18.6	20.6	23.6	26.9	28.1	28.9	26.6	22.9	18.2	15.7
		MCWB	12.9	12.4	13.1	14.5	16.4	18.4	19.2	19.2	18.1	17.0	15.4	13.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	16.0	15.3	16.3	18.7	20.7	21.7	22.3	22.6	21.3	20.7	19.1	16.8
		MCDB	16.5	16.7	21.4	25.4	28.8	31.6	32.7	32.2	28.9	23.2	20.5	17.4
	2%	WB	15.1	14.6	15.2	16.9	19.0	20.7	21.4	21.5	20.5	19.9	17.9	16.1
		MCDB	15.8	16.2	18.8	23.3	26.2	29.2	30.3	29.6	26.8	22.5	19.3	16.9
	5%	WB	14.2	13.8	14.6	15.9	17.9	19.8	20.5	20.7	19.8	19.1	17.0	15.2
		MCDB	15.2	15.6	18.0	21.2	24.1	27.2	28.4	27.7	25.2	21.8	18.5	16.2
	10%	WB	13.3	13.0	14.0	15.0	17.0	18.9	19.7	20.1	19.2	18.4	16.0	14.2
		MCDB	14.6	14.9	17.4	19.2	22.1	25.6	26.7	26.7	23.9	21.1	17.6	15.4
Mean Daily Temperature Range	5% DB	MDBR	5.7	6.5	7.4	7.6	8.2	9.2	9.8	10.1	8.8	7.2	5.9	5.5
		MCDBR	6.0	7.8	10.9	11.2	12.4	13.1	13.7	13.6	12.3	10.3	6.8	5.4
		MCWBR	4.1	4.4	5.2	5.5	5.8	5.6	5.3	5.3	4.9	4.4	4.0	4.0
	5% WB	MCDBR	4.6	6.1	8.4	9.9	10.5	12.0	12.0	11.3	9.5	7.0	5.4	4.7
		MCWBR	4.0	4.1	4.6	5.1	5.4	5.5	5.1	4.8	4.2	3.8	3.9	4.1
Clear-Sky Solar Irradiance	taub	0.326	0.339	0.368	0.385	0.393	0.397	0.382	0.395	0.394	0.360	0.336	0.327	
	taud	2.517	2.476	2.405	2.351	2.359	2.366	2.401	2.374	2.389	2.503	2.528	2.516	
	Ebn at Noon	851	886	890	892	888	881	892	872	851	850	832	821	
	Edh at Noon	77	91	108	121	123	122	117	117	109	88	76	73	
All-Sky Solar Radiation	RadAvg	2.24	3.24	4.40	5.66	6.79	7.34	7.48	6.79	5.31	3.65	2.50	2.02	
	RadStd	0.28	0.38	0.47	0.46	0.45	0.38	0.29	0.26	0.34	0.30	0.31	0.23	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	+0.24	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+102
(47)	Regional (15 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page